

EmbeddedAD: Efficient Anomaly Detection on SoCs via Vector-Quantized Reverse Distillation

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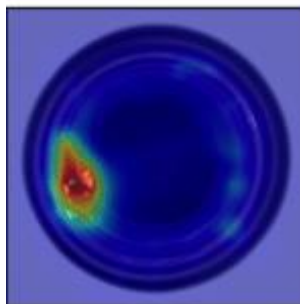
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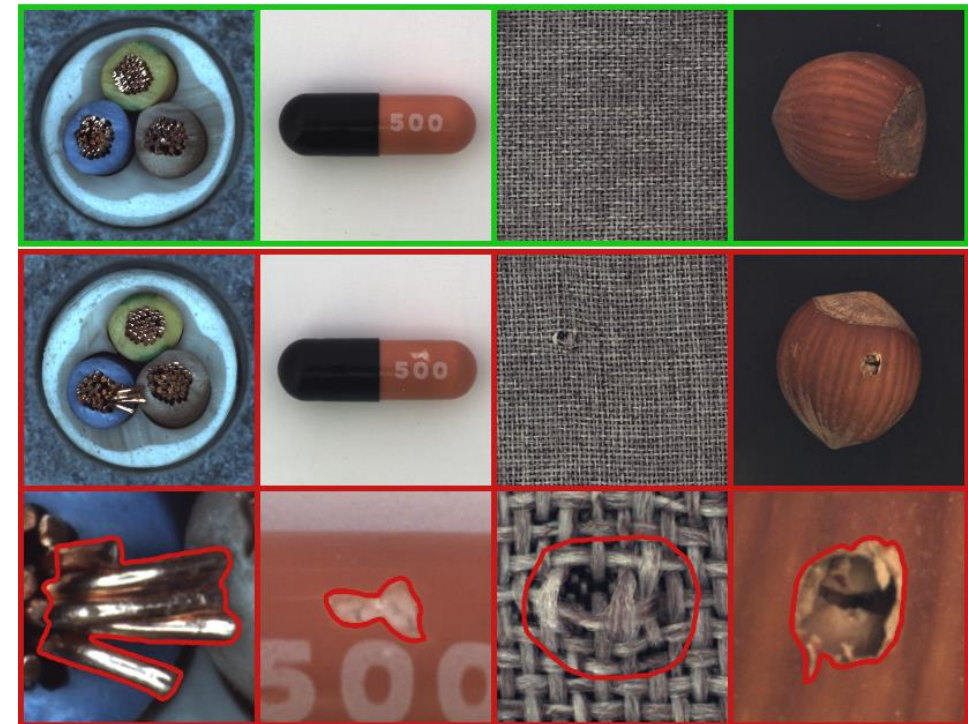
Identifying rare items, events, or observations that differ from recorded data.

Training AD models:

- Train model exclusively on normal data to learn a “state of normality”.
- Model can then detect anomalies as deviations from the learned state.
- AD models produce heat maps of anomaly scores at the pixel level, which makes localization possible.



Anomaly Map



Bergmann, Paul, et al. "MVTec AD—A comprehensive real-world dataset for unsupervised anomaly detection." 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2019.

Benchmark on Embedded SoC

- Existing AD models demonstrate excellent performance in image and pixel-level detection on the MVTec AD¹ benchmark.
- However, their inference latency on SoCs is relatively high (2-3 FPS) because they are designed and optimized for GPU execution.

→ **Goal of this work:** Design an AD model built around efficiency-first principles and optimized for execution on SoCs.

→ **Reverse Distillation (RD4AD)** as a starting point:
The model offers a balanced trade-off between inference latency and AD detection accuracy.

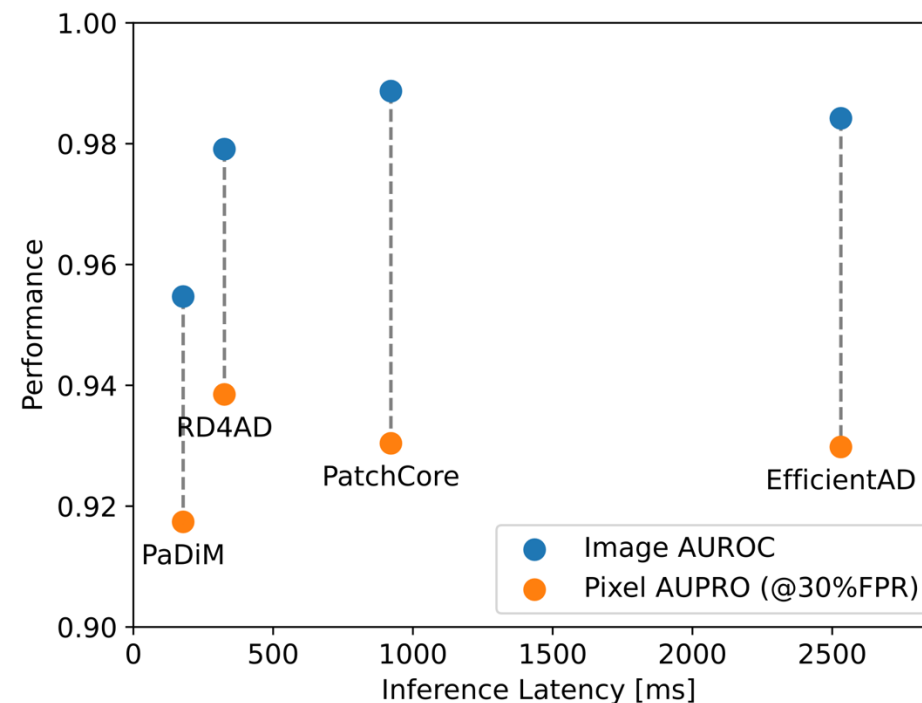
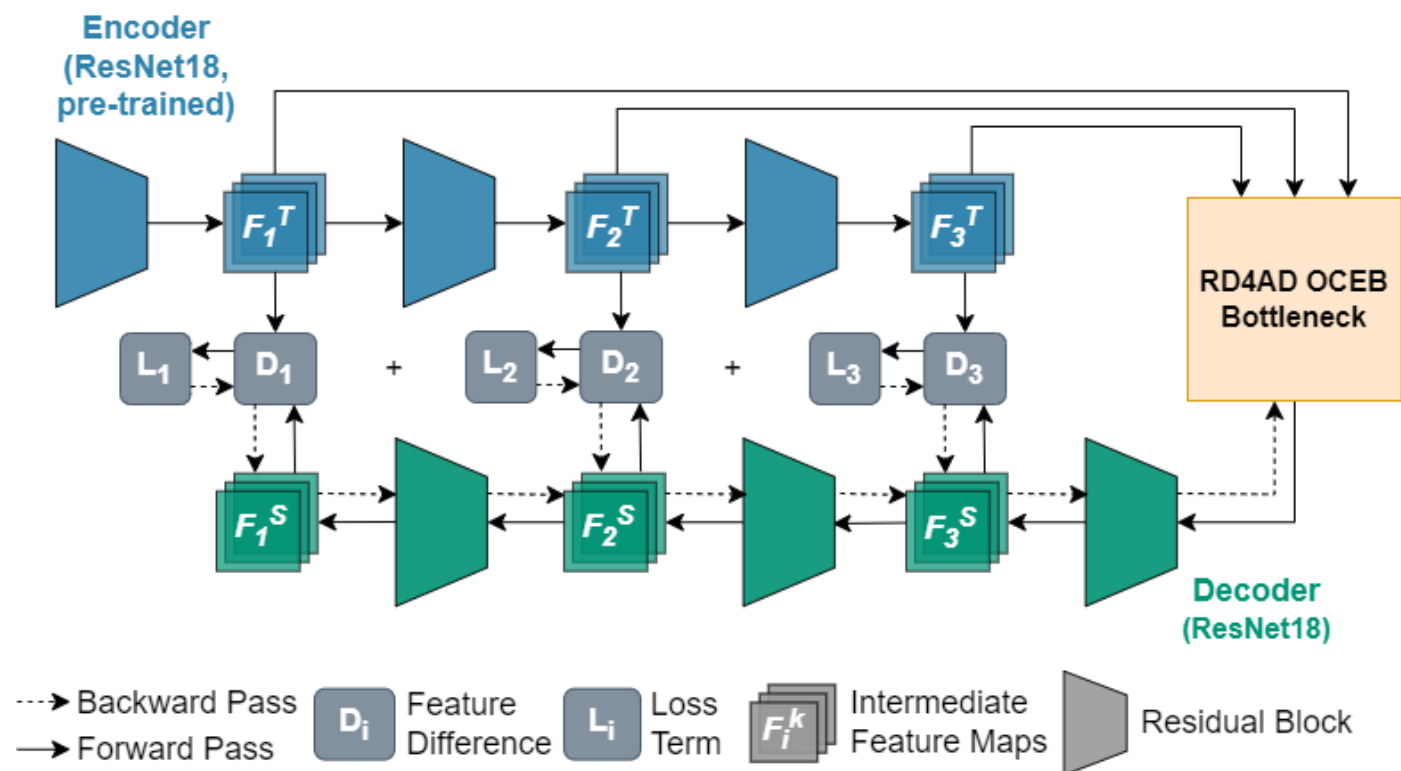


Image and pixel level accuracy on the MVTec AD benchmark vs. inference latency for different state-of-the-art AD models on a Raspberry Pi 5 using onnxruntime and all CPU cores.

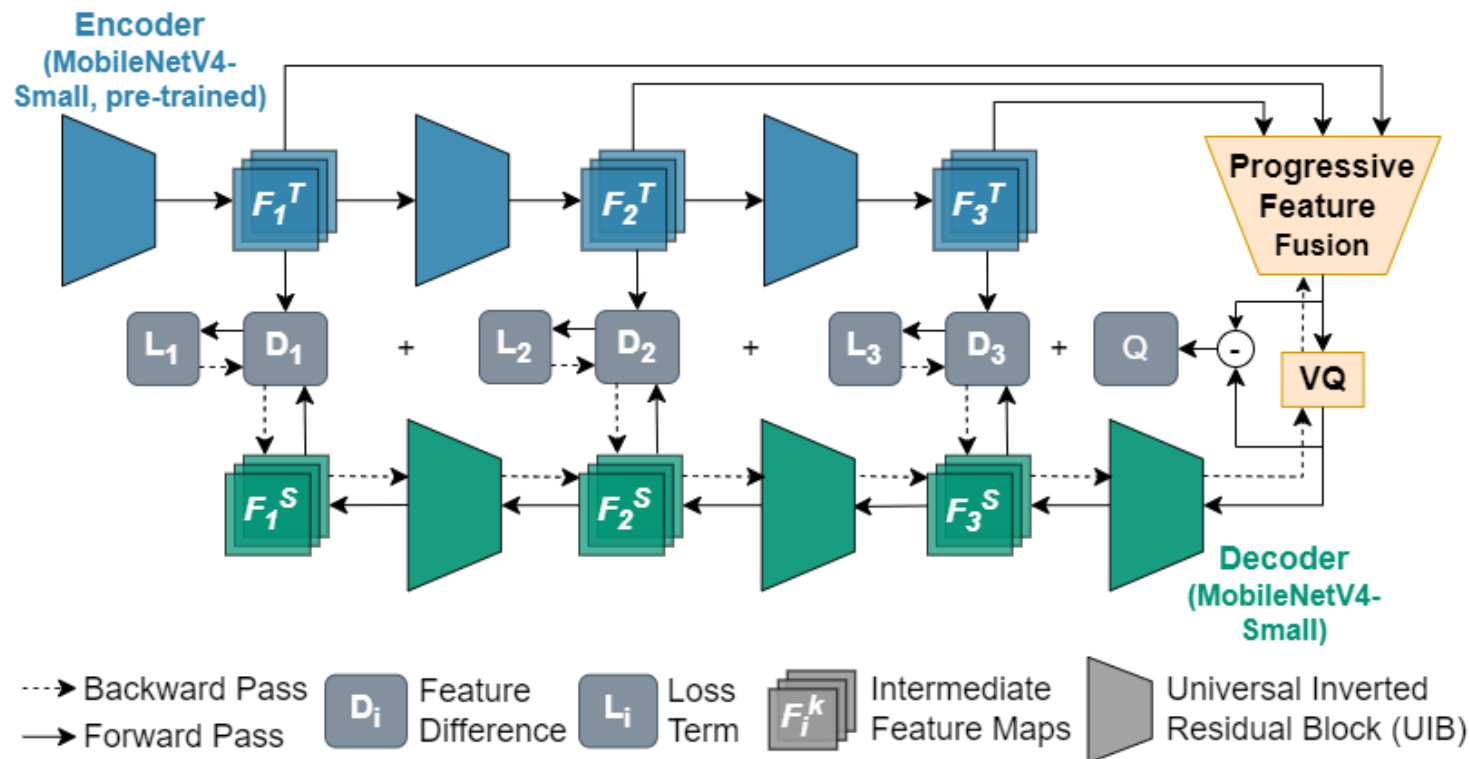
1. Bergmann, Paul, et al. "MVTec AD—A comprehensive real-world dataset for unsupervised anomaly detection." 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2019.

Efficient Reverse Distillation for Anomaly Detection



Backbone	Number of Parameters			
	Encoder	Decoder	Latent Block	Total
ResNet18 ¹	2.8 M	3.7 M	12.2 M	18.7 M

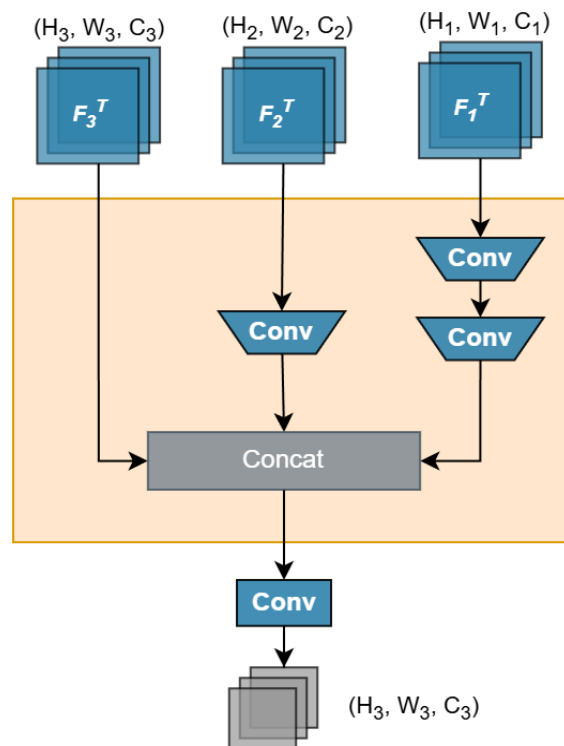
1. Deng, Hanqiu, and Xingyu Li. "Anomaly detection via reverse distillation from one-class embedding." 2022 IEEE/CVF conference on computer vision and pattern recognition (CVPR). IEEE, 2022.
2. He, Kaiming, et al. "Deep residual learning for image recognition." Proceedings of the IEEE conference on computer vision and pattern recognition. 2016.



Backbone	Number of Parameters			
	Encoder	Decoder	Latent Block	Total
MobileNetV4 Small ¹	0.3 M	0.5 M	0.2 M	1.0 M

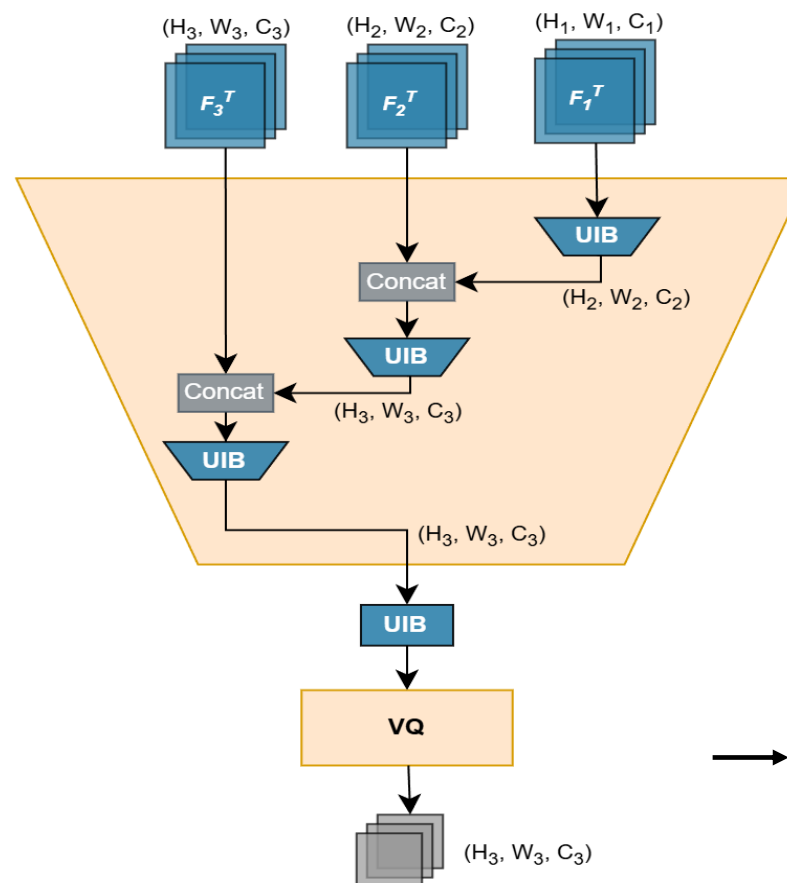
1. Qin, Danfeng, et al. "MobileNetV4: Universal models for the mobile ecosystem." *European conference on computer vision*. Cham: Springer Nature Switzerland, 2024.

RD4AD: One-Class Bottleneck Embedding (OCBE)



→ 12.2M Parameters

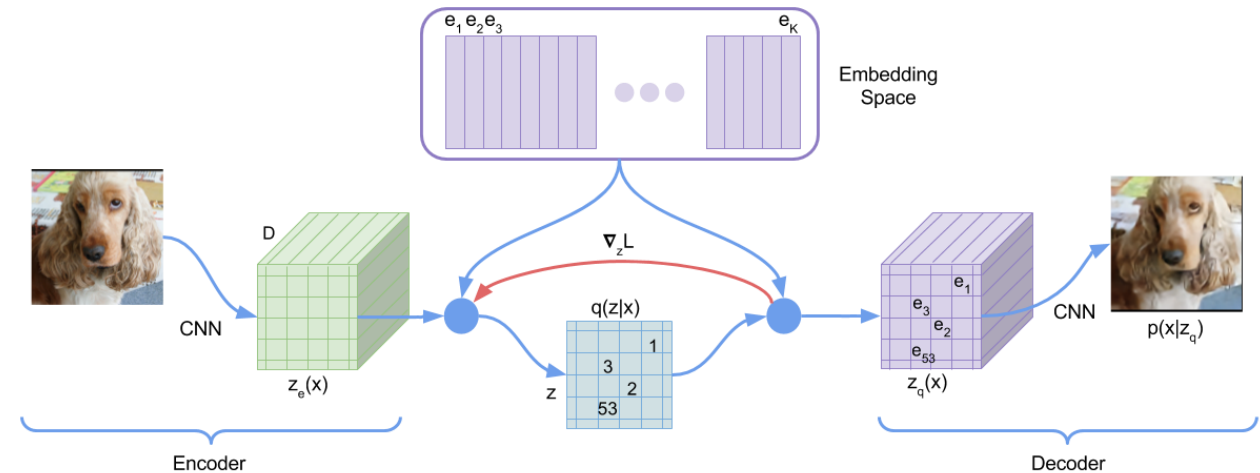
VQ-RD: Vector-Quantized Progressive Feature Fusion (VQ-PFF)



→ 0.2M Parameters

Discretized Latent Space

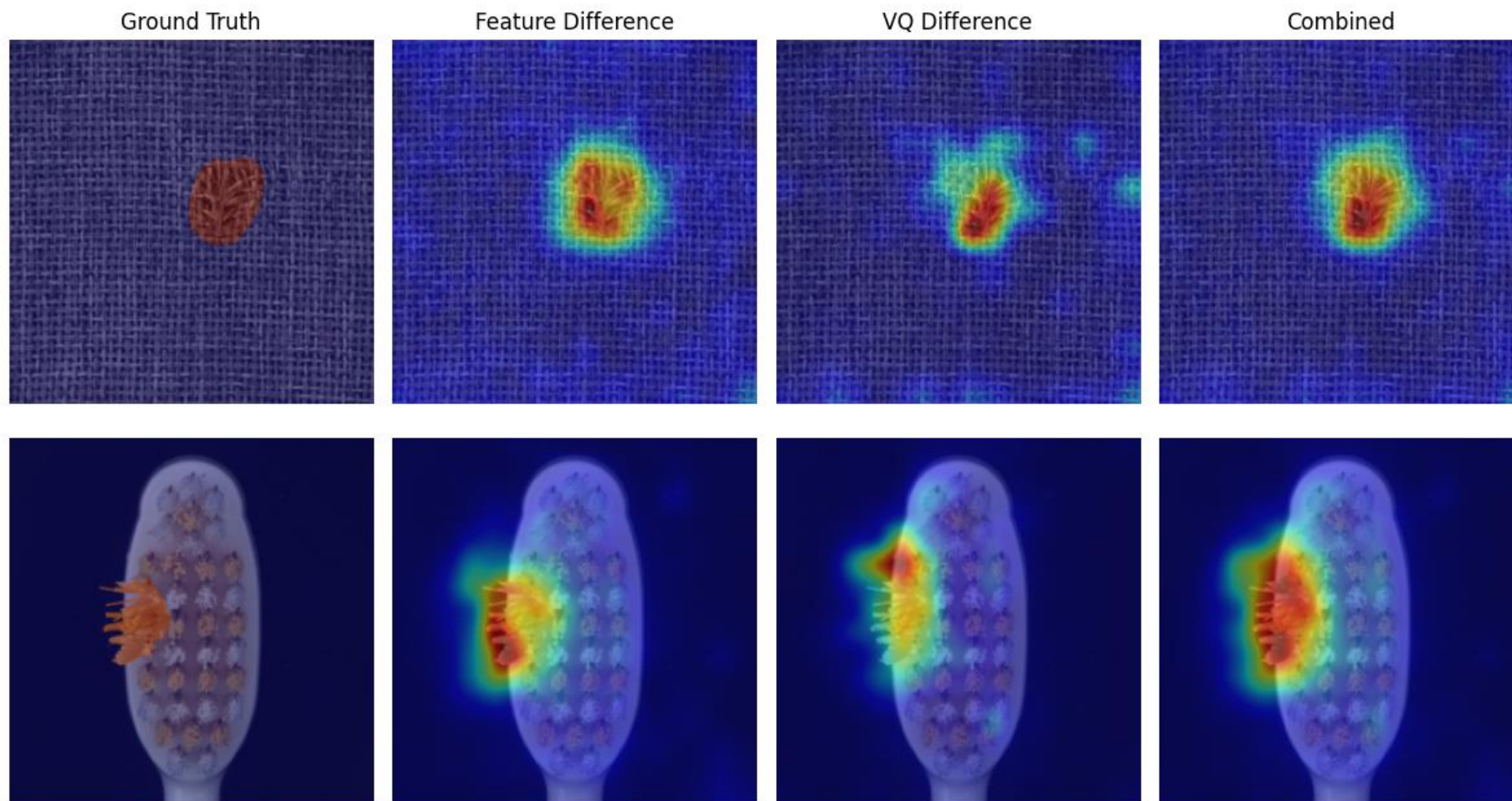
- Replaces continuous, probabilistic latent distribution of VAEs with a fixed discrete set of learned prototype vectors (codebook).
- Encoder outputs are quantized to the nearest vectors in the codebook.
- Can be thought of as latent-space feature clustering with codebook vectors representing cluster centroids.
- Distance to codebook vectors can be used as an indicator on how “anomalous” a sample is.



Van Den Oord, Aaron, and Oriol Vinyals. "Neural discrete representation learning." Advances in neural information processing systems (2017).

Efficient Reverse Distillation for Anomaly Detection

Combining Feature Similarity & Quantization Error



The Double Distillation Training Method

Limitations

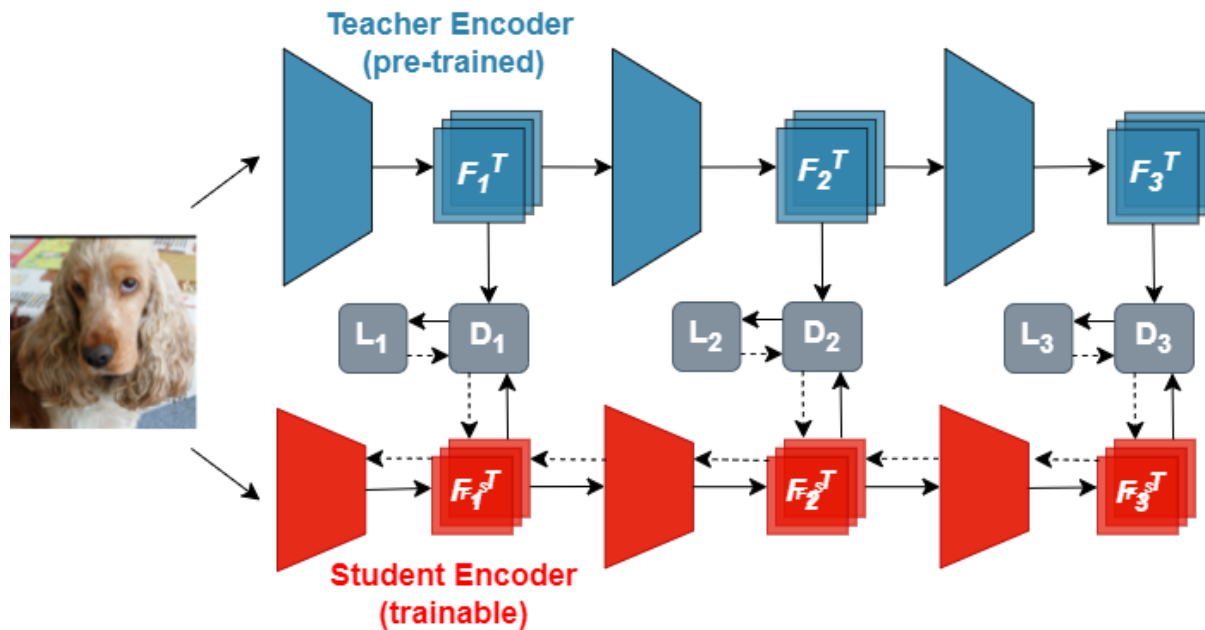
1. Models are dependent on existing **pre-trained** ImageNet DNN architectures because the encoder is not trained as part of the reverse distillation training method (e.g., ResNet18, MobileNetV4, ...).
2. Decoder must match the encoder to allow comparison of the intermediate feature maps.

→ **No direct control over the underlying DNN design**

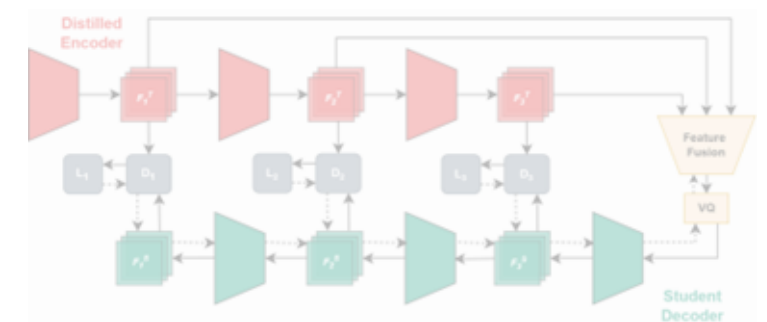
The Double Distillation Method

Enable Architectural Flexibility with Knowledge Distillation

STEP 1: Distill pretrained teacher into custom lightweight encoder



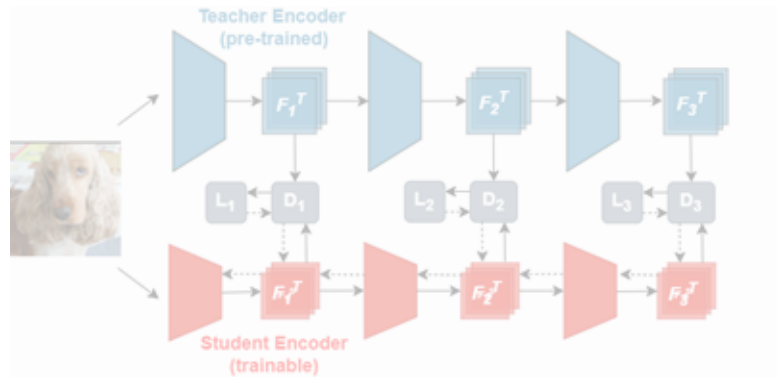
STEP 2: Use distilled custom encoder in the usual RD4AD-style training



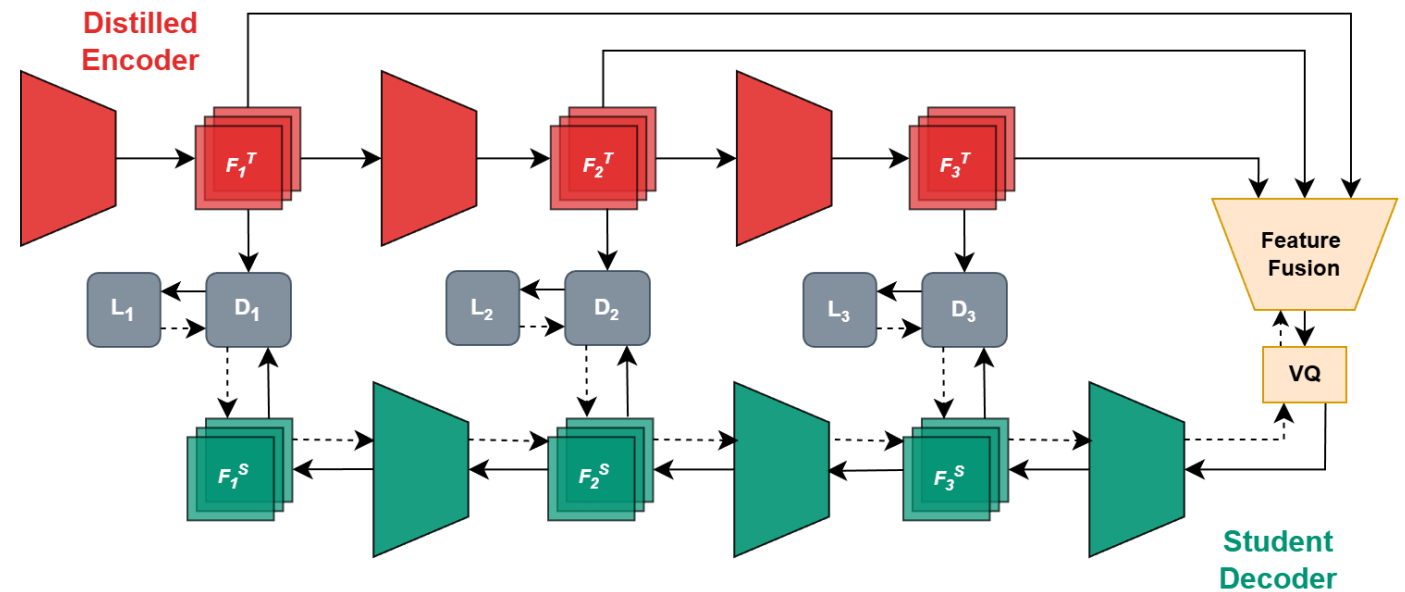
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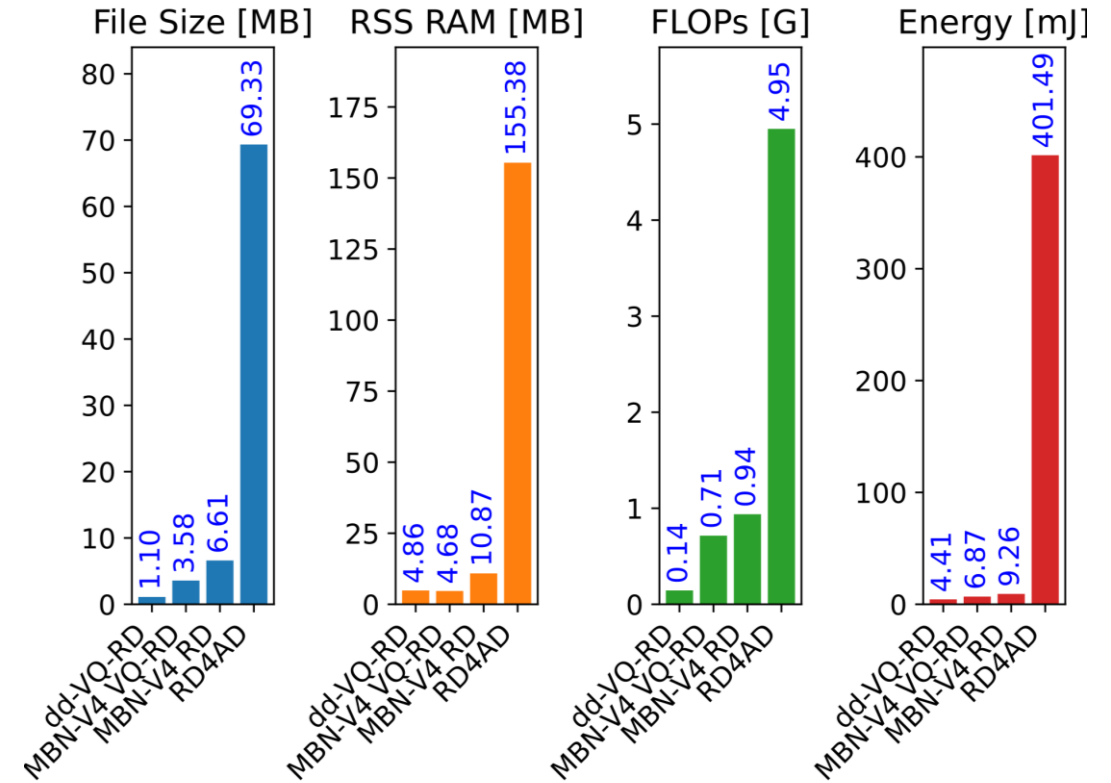
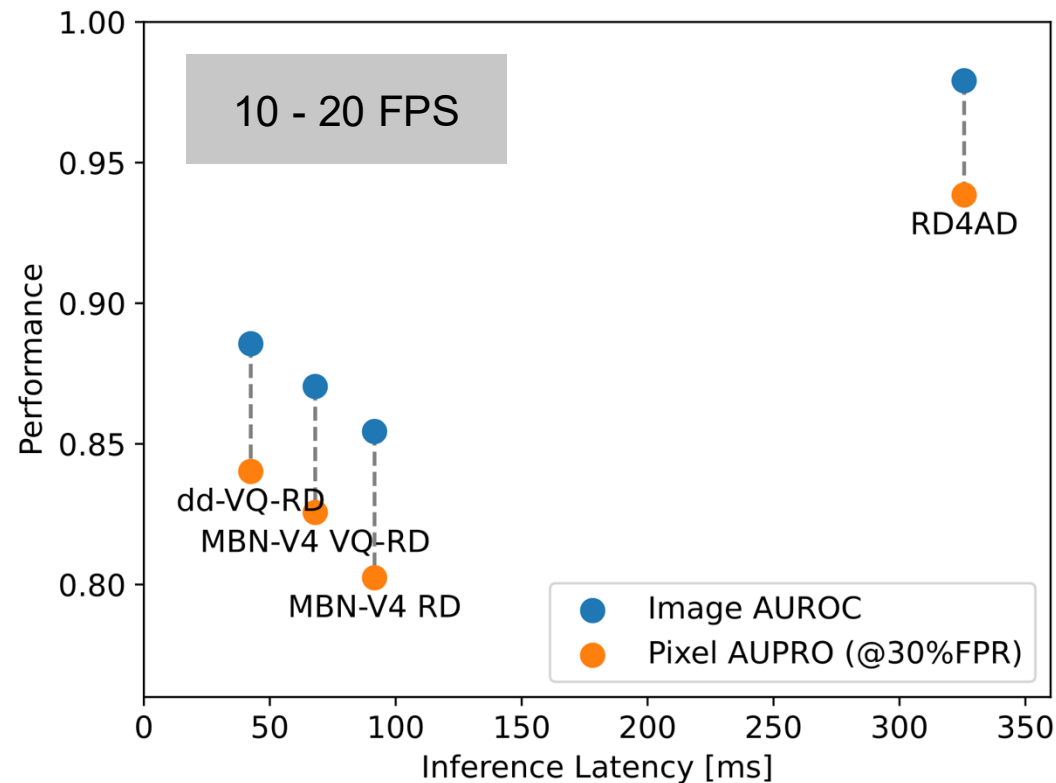
STEP 2: Use student encoder in the regular AD training



Evaluation

Evaluation

EmbeddedAD versus RD4AD



Left: Image and pixel-level accuracy on the MVTec AD benchmark versus inference latency for various EmbeddedAD models, compared to the RD4AD baseline, on a Raspberry Pi 5, using onnxruntime and all CPU cores. *Right:* A comparison of the resource requirements of the evaluated models. Energy was measured externally using a Joulescope power meter.



- Develop an efficient DNN architecture suitable for AD on embedded SoCs.
- Minimize degradation in image and pixel level AD precision of the efficient architecture.
- Make the architecture flexible and adaptable to the constraints of different SoCs.



- On a Raspberry Pi 5, EmbeddedAD achieves a 7.7x speedup at the cost of only a small degradation in AD performance on the MVTec AD dataset (−7% AUROC and AUPRO).
- The double distillation methods allows to freely adapt the encoder and decoder of Embedded AD without being limited to pre-trained architectures.

Thank you for you attention!